

# Junior Automation & Controls Technician

College Edition Competency Syllabus - Eight Integrated Modules

| PROGRAMME INFORMATION |                                                             |
|-----------------------|-------------------------------------------------------------|
| PROGRAMME WINDOW      | 7 September 2026 - 28 February 2027                         |
| CORE LOAD             | 25 weeks   19 hours/week   approximately 475 hours          |
| WEEKLY RHYTHM         | Monday-Friday study   Saturday laboratory or active project |
| PORTFOLIO             | Exactly three progressive low-voltage projects              |

## PURPOSE

A structured path from electrical fundamentals through PLCs, instrumentation, process control, SCADA, networking, troubleshooting, commissioning and engineering documentation.

## Prepared for Emmanuel Januarie

College-style personal learning programme - not an accredited qualification

# Course overview

This personal programme supports entry-level Junior Automation Technician, Junior Controls Technician, PLC Technician, Instrumentation & Controls Technician and E&I Technician roles. It combines theory, worked applications, simulation, safe hobby-scale physical work and three substantial portfolio projects.

## LEARNING PRINCIPLE

Study for usable understanding, not for a Sunday score. Weekday retrieval and application strengthen memory in context; Saturday proves the week's learning through an accurate laboratory or advances the active portfolio project. Sunday is optional light consolidation or rest.

## Fixed learning rhythm

| Day           | Duration           | Purpose                                                                                                  |
|---------------|--------------------|----------------------------------------------------------------------------------------------------------|
| Monday-Friday | 3 h/day            | One topic per day: retrieval, focused learning, worked examples, application, error log and next action. |
| Saturday      | 4 h                | Weeks 1-4: topic-matched laboratory. Weeks 5-25: active project only; no separate laboratory.            |
| Sunday        | Optional 30-60 min | Light review, filing or preview only. No test, score, submission deadline or new topic.                  |

## Programme outcomes

- [ ] Trace an electrical or instrument signal from process/field device through wiring, I/O, PLC logic, HMI and actuator.
- [ ] Create and troubleshoot maintainable Ladder Logic with modes, sequences, permissives, interlocks and recoverable faults.
- [ ] Explain and simulate analog instrumentation, 4-20 mA, scaling, calibration, PV/SP/feedback and PID fundamentals.
- [ ] Create HMI/SCADA screens, alarm/trend evidence and basic Ethernet/Modbus/OPC UA diagnostics.
- [ ] Produce coordinated P&IDs, schematics, loop drawings, registers, narratives, matrices, test sheets and reports.
- [ ] Work honestly within low-voltage DIY/simulation boundaries and distinguish training evidence from industrial authorization.

# Curriculum map and workload

| Module | Title                                             | Dates                     | Target                        | Core hours |
|--------|---------------------------------------------------|---------------------------|-------------------------------|------------|
| M1     | Electrical Fundamentals                           | 07 Sep 2026 - 04 Oct 2026 | Fundamentals                  | 76 h       |
| M2     | PLC & Automation                                  | 05 Oct 2026 - 15 Nov 2026 | Developing                    | 114 h      |
| M3     | Instrumentation                                   | 16 Nov 2026 - 13 Dec 2026 | Developing                    | 76 h       |
| M4     | Process Control                                   | 14 Dec 2026 - 27 Dec 2026 | Fundamentals                  | 38 h       |
| M5     | SCADA / HMI                                       | 28 Dec 2026 - 10 Jan 2027 | Developing                    | 38 h       |
| M6     | Industrial Networking                             | 11 Jan 2027 - 24 Jan 2027 | Working knowledge             | 38 h       |
| M7     | Engineering Documentation                         | 25 Jan 2027 - 07 Feb 2027 | Developing                    | 38 h       |
| M8     | Safety Awareness, Commissioning & Troubleshooting | 08 Feb 2027 - 28 Feb 2027 | Fundamentals with integration | 57 h       |

## PORTFOLIO WINDOWS

Project 1: 5 Oct-15 Nov 2026. Project 2: 16 Nov-27 Dec 2026. Project 3 flagship: 28 Dec 2026-28 Feb 2027. During these windows, Saturday is reserved entirely for the active project.

## Skill coverage

### TECHNICAL

- ☐ Electrical and motor control
- ☐ PLC architecture and programming
- ☐ Instrumentation and 4-20 mA
- ☐ Process control and PID
- ☐ HMI/SCADA and historian
- ☐ Ethernet and industrial protocols

### TECHNICIAN PRACTICE

- ☐ Schematics and P&IDs
- ☐ I/O and loop checks
- ☐ Fault isolation
- ☐ FAT/SAT-style testing
- ☐ Change and evidence control
- ☐ Safe-work awareness

# Learning, submission and evidence policies

## EVIDENCE LABELS

- ☐ Studied - explain in own words
- ☐ Simulated - software/hobby model
- ☐ Implemented - built/configured
- ☐ Tested - expected versus actual
- ☐ Documented - revision controlled
- ☐ Industrial claim - never implied

## FILE DISCIPLINE

- ☐ Project-document-revision names
- ☐ One evidence index per project
- ☐ Keep source and exported PDF
- ☐ Record software/version used
- ☐ Mark unperformed work NOT TESTED
- ☐ Back up before major changes

## Daily comprehension approach

Retrieval is brief and low-stakes: explain yesterday's idea, then use the answer immediately. Worked examples, application problems and error correction are learning activities, not examinations.

## Saturday rule

Weeks 1-4 use only the laboratory printed for that week. From Week 5 onward, do not add a separate laboratory: focus on the active project for the full four hours and choose the next incomplete section from your own design and build plan.

## Sunday boundary

Sunday may be rest or 30-60 minutes of light review, file organization, error-log review and preview. There is no Sunday test, formal assessment, homework submission or new topic.

## SAFETY AND HONESTY

Keep physical work at safe low voltage. Hobby components and DIY structures may represent industrial functions, but do not DIY mains, safety-critical, pressure-rated or hazardous-area equipment. State what is real, DIY-built, simulated or software-based.

# M1 - Electrical Fundamentals

| TARGET LEVEL | DATES                     | PLANNED CORE LOAD |
|--------------|---------------------------|-------------------|
| Fundamentals | 07 Sep 2026 - 04 Oct 2026 | 76 hours          |

## MODULE PURPOSE

Develop safe low-voltage circuit reasoning, measurement habits, component recognition, schematic tracing and motor/VFD awareness before PLC work.

## Module competencies

- [ ] Explain electrical quantities and units
- [ ] Rearrange  $V=IR$
- [ ] Calculate  $P=VI$
- [ ] Distinguish AC from DC
- [ ] Solve series/parallel networks
- [ ] Select safe meter methods
- [ ] Recognize control components
- [ ] Trace a holding circuit

## Module activities

- [ ] Calculate and annotate DC circuits
- [ ] Build breadboard LED/resistor circuits
- [ ] Trace relay/contactors control paths
- [ ] Simulate motor/VFD interfaces and faults

## Weekly sequence

| Programme week | Dates           | Theme          | Primary scope                                                                |
|----------------|-----------------|----------------|------------------------------------------------------------------------------|
| Week 1         | 07 Sep - 13 Sep | Electrical I   | Voltage, current, resistance, Ohm's Law, power and DC circuits               |
| Week 2         | 14 Sep - 20 Sep | Electrical II  | AC/DC, series and parallel circuits, measurement and multimeter fundamentals |
| Week 3         | 21 Sep - 27 Sep | Electrical III | Relays, contactors, circuit protection and industrial control circuits       |
| Week 4         | 28 Sep - 04 Oct | Electrical IV  | Motors, three-phase and VFD fundamentals plus basic troubleshooting          |

# M1 Module Week 1 - Programme Week 1

| WEEKLY FOCUS     |                                                                |
|------------------|----------------------------------------------------------------|
| Dates            | 07 Sep - 13 Sep 2026                                           |
| Theme            | Electrical I                                                   |
| Primary scope    | Voltage, current, resistance, Ohm's Law, power and DC circuits |
| Portfolio window | Not active - Saturday uses the week's laboratory               |

## Weekly learning outcomes

[ ] Explain electrical quantities and units

[ ] Rearrange  $V=IR$

[ ] Calculate  $P=VI$

## Monday-Sunday execution plan

| Day           | Time                  | Mode                             | Required study and application                                                                                                                                                                      |
|---------------|-----------------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon<br>07 Sep | 3 h                   | Voltage                          | Study focus: Voltage. Application: A battery measures 9 V. State the potential difference and identify positive and negative reference points.                                                      |
| Tue<br>08 Sep | 3 h                   | Current                          | Study focus: Current. Application: A branch carries 0.025 A. Convert to mA and show where an ammeter belongs.                                                                                       |
| Wed<br>09 Sep | 3 h                   | Resistance                       | Study focus: Resistance. Application: Compare 220 ohm and 1 kohm loads on the same source; predict which draws less current.                                                                        |
| Thu<br>10 Sep | 3 h                   | Ohm's Law                        | Study focus: Ohm's Law. Application: Calculate voltage for 15 mA through 330 ohms, then calculate current at 9 V.                                                                                   |
| Fri<br>11 Sep | 3 h                   | Electrical power and DC circuits | Study focus: Electrical power and DC circuits. Application: Calculate power for 9 V at 20 mA and select a supply with reasonable margin.                                                            |
| Sat<br>12 Sep | 4 h                   | Weekly laboratory                | Build a battery-resistor-LED circuit on the breadboard. Predict current, verify polarity, observe normal/open-circuit behavior and record expected versus actual results. Do not short the battery. |
| Sun<br>13 Sep | Optional<br>30-60 min | Light consolidation or rest      | No test and no new topic. If useful, review the error log, organize evidence and preview Monday; otherwise rest.                                                                                    |

### SESSION STRUCTURE

Monday-Friday: 15 min retrieval, 65 min focused learning, 10 min break, 45 min worked examples, 10 min break, 25 min application, 10 min error log and next action. Saturday: 4 hours using the laboratory or active-project rule shown above.

# M1 Module Week 2 - Programme Week 2

| WEEKLY FOCUS     |                                                                              |
|------------------|------------------------------------------------------------------------------|
| Dates            | 14 Sep - 20 Sep 2026                                                         |
| Theme            | Electrical II                                                                |
| Primary scope    | AC/DC, series and parallel circuits, measurement and multimeter fundamentals |
| Portfolio window | Not active - Saturday uses the week's laboratory                             |

## Weekly learning outcomes

[ ] Distinguish AC from DC

[ ] Solve series/parallel networks

[ ] Select safe meter methods

## Monday-Sunday execution plan

| Day           | Time                  | Mode                                           | Required study and application                                                                                                                                                                               |
|---------------|-----------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon<br>14 Sep | 3 h                   | AC and DC fundamentals                         | Study focus: AC and DC fundamentals. Application: Compare battery DC with mains AC and state why the home trainer remains low voltage.                                                                       |
| Tue<br>15 Sep | 3 h                   | Series circuits                                | Study focus: Series circuits. Application: Calculate total resistance and current for two series resistors on 9 V.                                                                                           |
| Wed<br>16 Sep | 3 h                   | Parallel circuits                              | Study focus: Parallel circuits. Application: Calculate branch and total current for two parallel resistors.                                                                                                  |
| Thu<br>17 Sep | 3 h                   | Voltage measurement                            | Study focus: Voltage measurement. Application: Create a safe voltage-measurement plan with reference point and expected range.                                                                               |
| Fri<br>18 Sep | 3 h                   | Current, resistance and continuity measurement | Study focus: Current, resistance and continuity measurement. Application: Choose the correct meter connection for voltage, current, resistance and continuity; identify unsafe energized resistance testing. |
| Sat<br>19 Sep | 4 h                   | Weekly laboratory                              | Build one series and one parallel LED/resistor circuit. Predict node voltages and branch behavior, compare brightness qualitatively, and document one open-branch fault. Use only battery-level voltage.     |
| Sun<br>20 Sep | Optional<br>30-60 min | Light consolidation or rest                    | No test and no new topic. If useful, review the error log, organize evidence and preview Monday; otherwise rest.                                                                                             |

### SESSION STRUCTURE

Monday-Friday: 15 min retrieval, 65 min focused learning, 10 min break, 45 min worked examples, 10 min break, 25 min application, 10 min error log and next action. Saturday: 4 hours using the laboratory or active-project rule shown above.

# M1 Module Week 3 - Programme Week 3

| WEEKLY FOCUS     |                                                                        |
|------------------|------------------------------------------------------------------------|
| Dates            | 21 Sep - 27 Sep 2026                                                   |
| Theme            | Electrical III                                                         |
| Primary scope    | Relays, contactors, circuit protection and industrial control circuits |
| Portfolio window | Not active - Saturday uses the week's laboratory                       |

## Weekly learning outcomes

[ ] Recognize control components

[ ] Trace a holding circuit

[ ] Explain protection boundaries

## Monday-Sunday execution plan

| Day           | Time                  | Mode                                  | Required study and application                                                                                                                                                                                     |
|---------------|-----------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon<br>21 Sep | 3 h                   | Relays and NO/NC contacts             | Study focus: Relays and NO/NC contacts. Application: Draw coil de-energized and energized contact states.                                                                                                          |
| Tue<br>22 Sep | 3 h                   | Contactors and auxiliary contacts     | Study focus: Contactors and auxiliary contacts. Application: Trace a contactor coil, main contacts and auxiliary feedback without using mains voltage.                                                             |
| Wed<br>23 Sep | 3 h                   | Fuses, breakers and overload concepts | Study focus: Fuses, breakers and overload concepts. Application: Match fuse, breaker and overload functions to faults they address.                                                                                |
| Thu<br>24 Sep | 3 h                   | Start/Stop and seal-in circuits       | Study focus: Start/Stop and seal-in circuits. Application: Create a Start/Stop truth table with NC Stop and holding contact.                                                                                       |
| Fri<br>25 Sep | 3 h                   | Electrical control schematics         | Study focus: Electrical control schematics. Application: Trace supply-to-load and supply-to-coil paths on a motor-control schematic.                                                                               |
| Sat<br>26 Sep | 4 h                   | Weekly laboratory                     | Create a breadboard relay/contactor simulation using LEDs as coil, motor and feedback indications. Use jumpers or buttons for Start, Stop and overload; prove seal-in logic with a software or manual state table. |
| Sun<br>27 Sep | Optional<br>30-60 min | Light consolidation or rest           | No test and no new topic. If useful, review the error log, organize evidence and preview Monday; otherwise rest.                                                                                                   |

### SESSION STRUCTURE

Monday-Friday: 15 min retrieval, 65 min focused learning, 10 min break, 45 min worked examples, 10 min break, 25 min application, 10 min error log and next action. Saturday: 4 hours using the laboratory or active-project rule shown above.

# M1 Module Week 4 - Programme Week 4

| WEEKLY FOCUS     |                                                                     |
|------------------|---------------------------------------------------------------------|
| Dates            | 28 Sep - 04 Oct 2026                                                |
| Theme            | Electrical IV                                                       |
| Primary scope    | Motors, three-phase and VFD fundamentals plus basic troubleshooting |
| Portfolio window | Not active - Saturday uses the week's laboratory                    |

## Weekly learning outcomes

[ ] Explain motor-control layers

[ ] Describe VFD interfaces

[ ] Troubleshoot systematically

## Monday-Sunday execution plan

| Day           | Time                  | Mode                             | Required study and application                                                                                                                                                                                    |
|---------------|-----------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon<br>28 Sep | 3 h                   | DC motor fundamentals            | Study focus: DC motor fundamentals. Application: Explain electrical-to-mechanical energy conversion in a small DC motor.                                                                                          |
| Tue<br>29 Sep | 3 h                   | Three-phase motor fundamentals   | Study focus: Three-phase motor fundamentals. Application: Interpret a sample three-phase nameplate at awareness level; do not wire mains equipment.                                                               |
| Wed<br>30 Sep | 3 h                   | Motor protection and overloads   | Study focus: Motor protection and overloads. Application: Separate short-circuit protection from overload protection.                                                                                             |
| Thu<br>01 Oct | 3 h                   | VFD fundamentals                 | Study focus: VFD fundamentals. Application: Map VFD ready, run, speed reference, running and fault signals.                                                                                                       |
| Fri<br>02 Oct | 3 h                   | Basic electrical troubleshooting | Study focus: Basic electrical troubleshooting. Application: Create a source-to-load troubleshooting sequence for a motor-will-not-run symptom.                                                                    |
| Sat<br>03 Oct | 4 h                   | Weekly laboratory                | Simulate a motor/VFD interface using LEDs for Ready, Run and Fault, a jumper for Run command and a state table for speed reference. Inject missing-ready and overload-open faults; record diagnosis and recovery. |
| Sun<br>04 Oct | Optional<br>30-60 min | Light consolidation or rest      | No test and no new topic. If useful, review the error log, organize evidence and preview Monday; otherwise rest.                                                                                                  |

### SESSION STRUCTURE

Monday-Friday: 15 min retrieval, 65 min focused learning, 10 min break, 45 min worked examples, 10 min break, 25 min application, 10 min error log and next action. Saturday: 4 hours using the laboratory or active-project rule shown above.

# M1 applied technician case study

## Conveyor motor indication will not start

### SITUATION

The low-voltage trainer has a Start request, but the motor indicator remains off. Supply polarity, Stop state, overload representation, holding path and output interface must be checked without guessing.

### Available evidence

- ☐ Schematic
- ☐ Expected voltages/states
- ☐ Measurement or simulation observations
- ☐ Corrected fault record

### Technician reasoning tasks

- ☐ Trace source to load
- ☐ Identify the first abnormal boundary
- ☐ State the safest confirming check
- ☐ Verify normal, fault and recovery behavior

### Required case-study record

| Record section              | What to capture                                                |
|-----------------------------|----------------------------------------------------------------|
| Symptom                     | Observable behavior without premature diagnosis                |
| Initial checks              | Safe, non-invasive checks in logical order                     |
| Evidence                    | Readings, states, screenshots or traces                        |
| Isolation                   | Boundary where expected and actual first differ                |
| Root cause                  | Verified cause, not a guess                                    |
| Correction and verification | Controlled change, normal/fault/recovery retest and limitation |

### LEARNING USE

This case study is an applied comprehension activity, not an examination. Use references, correct misconceptions and preserve the reasoning trail.

# M2 - PLC & Automation

| TARGET LEVEL | DATES                     | PLANNED CORE LOAD |
|--------------|---------------------------|-------------------|
| Developing   | 05 Oct 2026 - 15 Nov 2026 | 114 hours         |

## MODULE PURPOSE

Build PLC architecture, Ladder Logic, sequencing, mode control, protection, diagnostics and introductory Siemens/Rockwell awareness around Project 1.

## Module competencies

- ☐ Draw PLC architecture
- ☐ Explain the scan
- ☐ Write basic rungs
- ☐ Read rungs left to right
- ☐ Use timers/counters
- ☐ Control repeated events
- ☐ Separate permissive/interlock roles
- ☐ Define mode ownership

## Module activities

- ☐ Create I/O and tag lists
- ☐ Write and comment Ladder patterns
- ☐ Build modes, permissives and interlocks
- ☐ Run fault and recovery checks

## Weekly sequence

| Programme week | Dates           | Theme   | Primary scope                                                      |
|----------------|-----------------|---------|--------------------------------------------------------------------|
| Week 5         | 05 Oct - 11 Oct | PLC I   | PLC architecture, scan cycle, digital I/O, tags and Ladder basics  |
| Week 6         | 12 Oct - 18 Oct | PLC II  | Core Ladder instructions, latching, timers, counters and one-shots |
| Week 7         | 19 Oct - 25 Oct | PLC III | Modes, command ownership, permissives and interlocks               |
| Week 8         | 26 Oct - 01 Nov | PLC IV  | Sequencing, transitions, controlled restart and recovery           |
| Week 9         | 02 Nov - 08 Nov | PLC V   | Fault handling, alarms, trips, acknowledgement and diagnostics     |
| Week 10        | 09 Nov - 15 Nov | PLC VI  | Siemens fundamentals, online diagnostics and Rockwell awareness    |

# M2 Module Week 1 - Programme Week 5

| WEEKLY FOCUS     |                                                                   |
|------------------|-------------------------------------------------------------------|
| Dates            | 05 Oct - 11 Oct 2026                                              |
| Theme            | PLC I                                                             |
| Primary scope    | PLC architecture, scan cycle, digital I/O, tags and Ladder basics |
| Portfolio window | Project 1 active - Saturday is project focus only                 |

## Weekly learning outcomes

[ ] Draw PLC architecture

[ ] Explain the scan

[ ] Write basic rungs

## Monday-Sunday execution plan

| Day           | Time                  | Mode                               | Required study and application                                                                                                                    |
|---------------|-----------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon<br>05 Oct | 3 h                   | PLC architecture                   | Study focus: PLC architecture. Application: Draw power supply, CPU, DI, DO, communications and field devices.                                     |
| Tue<br>06 Oct | 3 h                   | PLC scan cycle                     | Study focus: PLC scan cycle. Application: Explain when an input change becomes visible to cyclic logic and the output update.                     |
| Wed<br>07 Oct | 3 h                   | Digital inputs                     | Study focus: Digital inputs. Application: Trace a hobby button or jumper from field state to a PLC-style input tag.                               |
| Thu<br>08 Oct | 3 h                   | Digital outputs                    | Study focus: Digital outputs. Application: Trace an output command through interface device to LED or motor representation.                       |
| Fri<br>09 Oct | 3 h                   | Tags and Ladder Logic fundamentals | Study focus: Tags and Ladder Logic fundamentals. Application: Write and test a basic Start/Stop rung with named tags.                             |
| Sat<br>10 Oct | 4 h                   | Project 1 focus                    | Focus on Project 1 only for 4 hours. Choose the next incomplete project section from your own design/build plan; no separate Saturday laboratory. |
| Sun<br>11 Oct | Optional<br>30-60 min | Light consolidation or rest        | No test and no new topic. If useful, review the error log, organize evidence and preview Monday; otherwise rest.                                  |

### SESSION STRUCTURE

Monday-Friday: 15 min retrieval, 65 min focused learning, 10 min break, 45 min worked examples, 10 min break, 25 min application, 10 min error log and next action. Saturday: 4 hours using the laboratory or active-project rule shown above.

# M2 Module Week 2 - Programme Week 6

| WEEKLY FOCUS     |                                                                    |
|------------------|--------------------------------------------------------------------|
| Dates            | 12 Oct - 18 Oct 2026                                               |
| Theme            | PLC II                                                             |
| Primary scope    | Core Ladder instructions, latching, timers, counters and one-shots |
| Portfolio window | Project 1 active - Saturday is project focus only                  |

## Weekly learning outcomes

- [ ] Read rungs left to right
- [ ] Use timers/counters
- [ ] Control repeated events

## Monday-Sunday execution plan

| Day           | Time                  | Mode                          | Required study and application                                                                                                                    |
|---------------|-----------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon<br>12 Oct | 3 h                   | Contacts and coils            | Study focus: Contacts and coils. Application: Predict rung truth for series and parallel contacts.                                                |
| Tue<br>13 Oct | 3 h                   | Seal-in and Set/Reset         | Study focus: Seal-in and Set/Reset. Application: Compare seal-in with Set/Reset and define safe reset behavior.                                   |
| Wed<br>14 Oct | 3 h                   | On-delay and off-delay timers | Study focus: On-delay and off-delay timers. Application: Create a delayed-start timing chart.                                                     |
| Thu<br>15 Oct | 3 h                   | Counters and comparisons      | Study focus: Counters and comparisons. Application: Build a count-to-five state table with reset.                                                 |
| Fri<br>16 Oct | 3 h                   | One-shots and edge detection  | Study focus: One-shots and edge detection. Application: Explain why a maintained sensor needs one-shot logic for one count per object.            |
| Sat<br>17 Oct | 4 h                   | Project 1 focus               | Focus on Project 1 only for 4 hours. Choose the next incomplete project section from your own design/build plan; no separate Saturday laboratory. |
| Sun<br>18 Oct | Optional<br>30-60 min | Light consolidation or rest   | No test and no new topic. If useful, review the error log, organize evidence and preview Monday; otherwise rest.                                  |

### SESSION STRUCTURE

Monday-Friday: 15 min retrieval, 65 min focused learning, 10 min break, 45 min worked examples, 10 min break, 25 min application, 10 min error log and next action. Saturday: 4 hours using the laboratory or active-project rule shown above.

# M2 Module Week 3 - Programme Week 7

| WEEKLY FOCUS     |                                                      |
|------------------|------------------------------------------------------|
| Dates            | 19 Oct - 25 Oct 2026                                 |
| Theme            | PLC III                                              |
| Primary scope    | Modes, command ownership, permissives and interlocks |
| Portfolio window | Project 1 active - Saturday is project focus only    |

## Weekly learning outcomes

- [ ] Separate permissive/interlock roles
- [ ] Define mode ownership
- [ ] Detect command-feedback mismatch

## Monday-Sunday execution plan

| Day           | Time                  | Mode                        | Required study and application                                                                                                                    |
|---------------|-----------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon<br>19 Oct | 3 h                   | Permissives                 | Study focus: Permissives. Application: Write the Boolean conditions required before a motor may start.                                            |
| Tue<br>20 Oct | 3 h                   | Interlocks                  | Study focus: Interlocks. Application: Explain how an interlock removes or prevents an unsafe command.                                             |
| Wed<br>21 Oct | 3 h                   | Auto/Manual control         | Study focus: Auto/Manual control. Application: Define what the operator may command in Manual versus Auto.                                        |
| Thu<br>22 Oct | 3 h                   | Local/Remote ownership      | Study focus: Local/Remote ownership. Application: Create an ownership table for Local and Remote modes.                                           |
| Fri<br>23 Oct | 3 h                   | Command versus feedback     | Study focus: Command versus feedback. Application: Design a timer that alarms when RunCmd is true but RunFb remains false.                        |
| Sat<br>24 Oct | 4 h                   | Project 1 focus             | Focus on Project 1 only for 4 hours. Choose the next incomplete project section from your own design/build plan; no separate Saturday laboratory. |
| Sun<br>25 Oct | Optional<br>30-60 min | Light consolidation or rest | No test and no new topic. If useful, review the error log, organize evidence and preview Monday; otherwise rest.                                  |

### SESSION STRUCTURE

Monday-Friday: 15 min retrieval, 65 min focused learning, 10 min break, 45 min worked examples, 10 min break, 25 min application, 10 min error log and next action. Saturday: 4 hours using the laboratory or active-project rule shown above.

# M2 Module Week 4 - Programme Week 8

| WEEKLY FOCUS     |                                                          |
|------------------|----------------------------------------------------------|
| Dates            | 26 Oct - 01 Nov 2026                                     |
| Theme            | PLC IV                                                   |
| Primary scope    | Sequencing, transitions, controlled restart and recovery |
| Portfolio window | Project 1 active - Saturday is project focus only        |

## Weekly learning outcomes

[ ] Design ordered sequences

[ ] Define transitions

[ ] Control recovery

## Monday-Sunday execution plan

| Day           | Time                  | Mode                            | Required study and application                                                                                                                    |
|---------------|-----------------------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon<br>26 Oct | 3 h                   | Sequence requirements           | Study focus: Sequence requirements. Application: Convert a conveyor description into numbered steps.                                              |
| Tue<br>27 Oct | 3 h                   | Steps and transitions           | Study focus: Steps and transitions. Application: Define measurable transition conditions between three steps.                                     |
| Wed<br>28 Oct | 3 h                   | State-machine fundamentals      | Study focus: State-machine fundamentals. Application: Draw a state diagram including Idle, Running, Complete and Faulted.                         |
| Thu<br>29 Oct | 3 h                   | Pause, abort and reset          | Study focus: Pause, abort and reset. Application: Specify pause, abort and reset behavior without skipping prerequisites.                         |
| Fri<br>30 Oct | 3 h                   | Power-loss and restart behavior | Study focus: Power-loss and restart behavior. Application: Decide what state must occur after power restoration and justify it.                   |
| Sat<br>31 Oct | 4 h                   | Project 1 focus                 | Focus on Project 1 only for 4 hours. Choose the next incomplete project section from your own design/build plan; no separate Saturday laboratory. |
| Sun<br>01 Nov | Optional<br>30-60 min | Light consolidation or rest     | No test and no new topic. If useful, review the error log, organize evidence and preview Monday; otherwise rest.                                  |

### SESSION STRUCTURE

Monday-Friday: 15 min retrieval, 65 min focused learning, 10 min break, 45 min worked examples, 10 min break, 25 min application, 10 min error log and next action. Saturday: 4 hours using the laboratory or active-project rule shown above.

# M2 Module Week 5 - Programme Week 9

| WEEKLY FOCUS     |                                                                |
|------------------|----------------------------------------------------------------|
| Dates            | 02 Nov - 08 Nov 2026                                           |
| Theme            | PLC V                                                          |
| Primary scope    | Fault handling, alarms, trips, acknowledgement and diagnostics |
| Portfolio window | Project 1 active - Saturday is project focus only              |

## Weekly learning outcomes

[ ] Design useful diagnostics

[ ] Handle faults predictably

[ ] Troubleshoot by layer

## Monday-Sunday execution plan

| Day           | Time                  | Mode                         | Required study and application                                                                                                                    |
|---------------|-----------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon<br>02 Nov | 3 h                   | Fault detection              | Study focus: Fault detection. Application: Define five detectable conveyor faults and their evidence.                                             |
| Tue<br>03 Nov | 3 h                   | Alarm versus trip            | Study focus: Alarm versus trip. Application: Classify warning, alarm and trip responses.                                                          |
| Wed<br>04 Nov | 3 h                   | Acknowledgement versus reset | Study focus: Acknowledgement versus reset. Application: Create an acknowledgement/reset state table.                                              |
| Thu<br>05 Nov | 3 h                   | First-out and fault memory   | Study focus: First-out and fault memory. Application: Preserve the first cause when several conditions follow a trip.                             |
| Fri<br>06 Nov | 3 h                   | PLC troubleshooting workflow | Study focus: PLC troubleshooting workflow. Application: Trace field device, wiring, I/O, logic, HMI and process layers before editing code.       |
| Sat<br>07 Nov | 4 h                   | Project 1 focus              | Focus on Project 1 only for 4 hours. Choose the next incomplete project section from your own design/build plan; no separate Saturday laboratory. |
| Sun<br>08 Nov | Optional<br>30-60 min | Light consolidation or rest  | No test and no new topic. If useful, review the error log, organize evidence and preview Monday; otherwise rest.                                  |

### SESSION STRUCTURE

Monday-Friday: 15 min retrieval, 65 min focused learning, 10 min break, 45 min worked examples, 10 min break, 25 min application, 10 min error log and next action. Saturday: 4 hours using the laboratory or active-project rule shown above.

# M2 Module Week 6 - Programme Week 10

| WEEKLY FOCUS     |                                                                 |
|------------------|-----------------------------------------------------------------|
| Dates            | 09 Nov - 15 Nov 2026                                            |
| Theme            | PLC VI                                                          |
| Primary scope    | Siemens fundamentals, online diagnostics and Rockwell awareness |
| Portfolio window | Project 1 active - Saturday is project focus only               |

## Weekly learning outcomes

- [ ] Navigate a vendor project
- [ ] Use diagnostics first
- [ ] Translate basic vendor concepts

## Monday-Sunday execution plan

| Day           | Time                  | Mode                              | Required study and application                                                                                                                    |
|---------------|-----------------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon<br>09 Nov | 3 h                   | TIA Portal project structure      | Study focus: TIA Portal project structure. Application: Identify hardware configuration, tags, blocks and watch tables in a TIA project.          |
| Tue<br>10 Nov | 3 h                   | S7-1200/S7-1500 fundamentals      | Study focus: S7-1200/S7-1500 fundamentals. Application: Compare compact and modular PLC concepts without claiming hardware experience.            |
| Wed<br>11 Nov | 3 h                   | Monitoring and diagnostic buffers | Study focus: Monitoring and diagnostic buffers. Application: Interpret a simulated diagnostic event before changing logic.                        |
| Thu<br>12 Nov | 3 h                   | Backup and change control         | Study focus: Backup and change control. Application: Create a versioned backup and change note.                                                   |
| Fri<br>13 Nov | 3 h                   | Allen-Bradley/Rockwell awareness  | Study focus: Allen-Bradley/Rockwell awareness. Application: Translate Siemens tag/block terminology into introductory Logix concepts.             |
| Sat<br>14 Nov | 4 h                   | Project 1 focus                   | Focus on Project 1 only for 4 hours. Choose the next incomplete project section from your own design/build plan; no separate Saturday laboratory. |
| Sun<br>15 Nov | Optional<br>30-60 min | Light consolidation or rest       | No test and no new topic. If useful, review the error log, organize evidence and preview Monday; otherwise rest.                                  |

### SESSION STRUCTURE

Monday-Friday: 15 min retrieval, 65 min focused learning, 10 min break, 45 min worked examples, 10 min break, 25 min application, 10 min error log and next action. Saturday: 4 hours using the laboratory or active-project rule shown above.

# M2 applied technician case study

## Conveyor stops during transfer

### SITUATION

The sequence is active, safety and overload conditions are healthy, but downstream-ready is false. The learner must distinguish field input, wiring, I/O, tag, logic and sequence causes.

### Available evidence

- ☐ I/O state table
- ☐ Logic screenshot
- ☐ Sequence state
- ☐ Fault timeline

### Technician reasoning tasks

- ☐ Confirm the active step
- ☐ Trace downstream-ready end to end
- ☐ Explain why code edits are not first
- ☐ Retest after correction

### Required case-study record

| Record section              | What to capture                                                |
|-----------------------------|----------------------------------------------------------------|
| Symptom                     | Observable behavior without premature diagnosis                |
| Initial checks              | Safe, non-invasive checks in logical order                     |
| Evidence                    | Readings, states, screenshots or traces                        |
| Isolation                   | Boundary where expected and actual first differ                |
| Root cause                  | Verified cause, not a guess                                    |
| Correction and verification | Controlled change, normal/fault/recovery retest and limitation |

### LEARNING USE

This case study is an applied comprehension activity, not an examination. Use references, correct misconceptions and preserve the reasoning trail.

# M3 - Instrumentation

| TARGET LEVEL | DATES                     | PLANNED CORE LOAD |
|--------------|---------------------------|-------------------|
| Developing   | 16 Nov 2026 - 13 Dec 2026 | 76 hours          |

## MODULE PURPOSE

Understand measurement chains, common process measurements, signal types, 4-20 mA, scaling, calibration records, valves and HART fundamentals.

## Module competencies

- ☐ Trace the measurement chain
- ☐ Distinguish signal types
- ☐ Explain live zero
- ☐ Compare temperature technologies
- ☐ Scale a signal
- ☐ Diagnose measurement faults
- ☐ Explain pressure reference
- ☐ Select level methods

## Module activities

- ☐ Build signal-chain drawings
- ☐ Calculate percent/current/engineering units
- ☐ Compare sensor technologies
- ☐ Prepare calibration and loop-check sheets

## Weekly sequence

| Programme week | Dates           | Theme               | Primary scope                                             |
|----------------|-----------------|---------------------|-----------------------------------------------------------|
| Week 11        | 16 Nov - 22 Nov | Instrumentation I   | Sensors, transmitters, digital/analog signals and 4-20 mA |
| Week 12        | 23 Nov - 29 Nov | Instrumentation II  | Temperature measurement, selection, scaling and faults    |
| Week 13        | 30 Nov - 06 Dec | Instrumentation III | Pressure and level measurement                            |
| Week 14        | 07 Dec - 13 Dec | Instrumentation IV  | Flow, calibration, valves, actuators and HART awareness   |

# M3 Module Week 1 - Programme Week 11

| WEEKLY FOCUS     |                                                           |
|------------------|-----------------------------------------------------------|
| Dates            | 16 Nov - 22 Nov 2026                                      |
| Theme            | Instrumentation I                                         |
| Primary scope    | Sensors, transmitters, digital/analog signals and 4-20 mA |
| Portfolio window | Project 2 active - Saturday is project focus only         |

## Weekly learning outcomes

[ ] Trace the measurement chain

[ ] Distinguish signal types

[ ] Explain live zero

## Monday-Sunday execution plan

| Day           | Time                  | Mode                        | Required study and application                                                                                                                    |
|---------------|-----------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon<br>16 Nov | 3 h                   | Sensors and transducers     | Study focus: Sensors and transducers. Application: Trace a physical variable from sensor to displayed tag.                                        |
| Tue<br>17 Nov | 3 h                   | Transmitters                | Study focus: Transmitters. Application: Explain why a transmitter standardizes the sensor response.                                               |
| Wed<br>18 Nov | 3 h                   | Digital instrument signals  | Study focus: Digital instrument signals. Application: Create a fail-safe level-switch truth table.                                                |
| Thu<br>19 Nov | 3 h                   | Analog instrument signals   | Study focus: Analog instrument signals. Application: Compare 0-10 V and 4-20 mA for noise and fault detection.                                    |
| Fri<br>20 Nov | 3 h                   | 4-20 mA and live zero       | Study focus: 4-20 mA and live zero. Application: Calculate percent span at 4, 8, 12, 16 and 20 mA.                                                |
| Sat<br>21 Nov | 4 h                   | Project 2 focus             | Focus on Project 2 only for 4 hours. Choose the next incomplete project section from your own design/build plan; no separate Saturday laboratory. |
| Sun<br>22 Nov | Optional<br>30-60 min | Light consolidation or rest | No test and no new topic. If useful, review the error log, organize evidence and preview Monday; otherwise rest.                                  |

### SESSION STRUCTURE

Monday-Friday: 15 min retrieval, 65 min focused learning, 10 min break, 45 min worked examples, 10 min break, 25 min application, 10 min error log and next action. Saturday: 4 hours using the laboratory or active-project rule shown above.

# M3 Module Week 2 - Programme Week 12

| WEEKLY FOCUS     |                                                        |
|------------------|--------------------------------------------------------|
| Dates            | 23 Nov - 29 Nov 2026                                   |
| Theme            | Instrumentation II                                     |
| Primary scope    | Temperature measurement, selection, scaling and faults |
| Portfolio window | Project 2 active - Saturday is project focus only      |

## Weekly learning outcomes

[ ] Compare temperature technologies

[ ] Scale a signal

[ ] Diagnose measurement faults

## Monday-Sunday execution plan

| Day           | Time                  | Mode                                      | Required study and application                                                                                                                    |
|---------------|-----------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon<br>23 Nov | 3 h                   | Temperature and units                     | Study focus: Temperature and units. Application: Convert between degrees C and K and define the measured range.                                   |
| Tue<br>24 Nov | 3 h                   | RTD fundamentals                          | Study focus: RTD fundamentals. Application: Explain RTD resistance change and 2/3/4-wire lead effects.                                            |
| Wed<br>25 Nov | 3 h                   | Thermocouple fundamentals                 | Study focus: Thermocouple fundamentals. Application: Explain thermocouple millivolts and cold-junction awareness.                                 |
| Thu<br>26 Nov | 3 h                   | Temperature transmitters and scaling      | Study focus: Temperature transmitters and scaling. Application: Scale 4-20 mA to 0-100 degrees C.                                                 |
| Fri<br>27 Nov | 3 h                   | Temperature faults and calibration checks | Study focus: Temperature faults and calibration checks. Application: Distinguish open sensor, wrong type, wiring error and actual process change. |
| Sat<br>28 Nov | 4 h                   | Project 2 focus                           | Focus on Project 2 only for 4 hours. Choose the next incomplete project section from your own design/build plan; no separate Saturday laboratory. |
| Sun<br>29 Nov | Optional<br>30-60 min | Light consolidation or rest               | No test and no new topic. If useful, review the error log, organize evidence and preview Monday; otherwise rest.                                  |

### SESSION STRUCTURE

Monday-Friday: 15 min retrieval, 65 min focused learning, 10 min break, 45 min worked examples, 10 min break, 25 min application, 10 min error log and next action. Saturday: 4 hours using the laboratory or active-project rule shown above.

# M3 Module Week 3 - Programme Week 13

| WEEKLY FOCUS     |                                                   |
|------------------|---------------------------------------------------|
| Dates            | 30 Nov - 06 Dec 2026                              |
| Theme            | Instrumentation III                               |
| Primary scope    | Pressure and level measurement                    |
| Portfolio window | Project 2 active - Saturday is project focus only |

## Weekly learning outcomes

[ ] Explain pressure reference

[ ] Select level methods

[ ] Troubleshoot by cause category

## Monday-Sunday execution plan

| Day           | Time                  | Mode                                   | Required study and application                                                                                                                    |
|---------------|-----------------------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon<br>30 Nov | 3 h                   | Pressure fundamentals                  | Study focus: Pressure fundamentals. Application: Differentiate gauge, absolute and differential pressure.                                         |
| Tue<br>01 Dec | 3 h                   | Pressure transmitter selection         | Study focus: Pressure transmitter selection. Application: Select a range and wetted-material assumptions for a small water system.                |
| Wed<br>02 Dec | 3 h                   | Level measurement technologies         | Study focus: Level measurement technologies. Application: Compare float, ultrasonic, hydrostatic and simulated level measurement.                 |
| Thu<br>03 Dec | 3 h                   | Level scaling and independent switches | Study focus: Level scaling and independent switches. Application: Scale a level input and define low-low/high-high switches.                      |
| Fri<br>04 Dec | 3 h                   | Pressure/level troubleshooting         | Study focus: Pressure/level troubleshooting. Application: Separate process, installation, wiring, configuration and device causes.                |
| Sat<br>05 Dec | 4 h                   | Project 2 focus                        | Focus on Project 2 only for 4 hours. Choose the next incomplete project section from your own design/build plan; no separate Saturday laboratory. |
| Sun<br>06 Dec | Optional<br>30-60 min | Light consolidation or rest            | No test and no new topic. If useful, review the error log, organize evidence and preview Monday; otherwise rest.                                  |

### SESSION STRUCTURE

Monday-Friday: 15 min retrieval, 65 min focused learning, 10 min break, 45 min worked examples, 10 min break, 25 min application, 10 min error log and next action. Saturday: 4 hours using the laboratory or active-project rule shown above.

# M3 Module Week 4 - Programme Week 14

| WEEKLY FOCUS     |                                                         |
|------------------|---------------------------------------------------------|
| Dates            | 07 Dec - 13 Dec 2026                                    |
| Theme            | Instrumentation IV                                      |
| Primary scope    | Flow, calibration, valves, actuators and HART awareness |
| Portfolio window | Project 2 active - Saturday is project focus only       |

## Weekly learning outcomes

- [ ] Scale flow signals
- [ ] Record calibration evidence
- [ ] Explain final elements and HART

## Monday-Sunday execution plan

| Day           | Time                  | Mode                            | Required study and application                                                                                                                    |
|---------------|-----------------------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon<br>07 Dec | 3 h                   | Flow measurement fundamentals   | Study focus: Flow measurement fundamentals. Application: Compare pulse, switch and analog flow signals.                                           |
| Tue<br>08 Dec | 3 h                   | Flow sensor signals and scaling | Study focus: Flow sensor signals and scaling. Application: Convert pulse frequency or 4-20 mA into engineering units.                             |
| Wed<br>09 Dec | 3 h                   | Calibration fundamentals        | Study focus: Calibration fundamentals. Application: Prepare three-point as-found/as-left records using simulation.                                |
| Thu<br>10 Dec | 3 h                   | Control valves and actuators    | Study focus: Control valves and actuators. Application: Define valve command, position feedback, fail position and travel fault.                  |
| Fri<br>11 Dec | 3 h                   | HART fundamentals               | Study focus: HART fundamentals. Application: Explain how HART information coexists with a 4-20 mA signal at awareness level.                      |
| Sat<br>12 Dec | 4 h                   | Project 2 focus                 | Focus on Project 2 only for 4 hours. Choose the next incomplete project section from your own design/build plan; no separate Saturday laboratory. |
| Sun<br>13 Dec | Optional<br>30-60 min | Light consolidation or rest     | No test and no new topic. If useful, review the error log, organize evidence and preview Monday; otherwise rest.                                  |

### SESSION STRUCTURE

Monday-Friday: 15 min retrieval, 65 min focused learning, 10 min break, 45 min worked examples, 10 min break, 25 min application, 10 min error log and next action. Saturday: 4 hours using the laboratory or active-project rule shown above.

# M3 applied technician case study

## Level indication is wrong after replacement

### SITUATION

A replacement level device is healthy but the displayed value is consistently incorrect. The configured range does not match the documented device range.

### Available evidence

- ☐ Device/simulated range
- ☐ Raw signal
- ☐ PLC scaling
- ☐ Displayed value and quality

### Technician reasoning tasks

- ☐ Calculate the expected value
- ☐ Identify configuration versus calibration
- ☐ Correct the documented parameter
- ☐ Verify low, midpoint and high points

### Required case-study record

| Record section              | What to capture                                                |
|-----------------------------|----------------------------------------------------------------|
| Symptom                     | Observable behavior without premature diagnosis                |
| Initial checks              | Safe, non-invasive checks in logical order                     |
| Evidence                    | Readings, states, screenshots or traces                        |
| Isolation                   | Boundary where expected and actual first differ                |
| Root cause                  | Verified cause, not a guess                                    |
| Correction and verification | Controlled change, normal/fault/recovery retest and limitation |

### LEARNING USE

This case study is an applied comprehension activity, not an examination. Use references, correct misconceptions and preserve the reasoning trail.

# M4 - Process Control

| TARGET LEVEL | DATES                     | PLANNED CORE LOAD |
|--------------|---------------------------|-------------------|
| Fundamentals | 14 Dec 2026 - 27 Dec 2026 | 38 hours          |

## MODULE PURPOSE

Connect measurement, control logic and final elements through PV, SP, MV, feedback, Manual/Auto, PID, alarms and protective concepts.

## Module competencies

- ☐ Label loop variables
- ☐ Explain feedback
- ☐ Compare Manual/Auto
- ☐ Explain PID terms
- ☐ Interpret trends
- ☐ Design protective responses

## Module activities

- ☐ Draw feedback loops
- ☐ Interpret response trends
- ☐ Write alarm responses
- ☐ Create cause/effect rows

## Weekly sequence

| Programme week | Dates           | Theme              | Primary scope                                                      |
|----------------|-----------------|--------------------|--------------------------------------------------------------------|
| Week 15        | 14 Dec - 20 Dec | Process Control I  | Process variables, PV/SP/MV, feedback and Manual/Automatic control |
| Week 16        | 21 Dec - 27 Dec | Process Control II | PID fundamentals, alarms, permissives, interlocks and ESD concepts |

# M4 Module Week 1 - Programme Week 15

| WEEKLY FOCUS     |                                                                    |
|------------------|--------------------------------------------------------------------|
| Dates            | 14 Dec - 20 Dec 2026                                               |
| Theme            | Process Control I                                                  |
| Primary scope    | Process variables, PV/SP/MV, feedback and Manual/Automatic control |
| Portfolio window | Project 2 active - Saturday is project focus only                  |

## Weekly learning outcomes

[ ] Label loop variables

[ ] Explain feedback

[ ] Compare Manual/Auto

## Monday-Sunday execution plan

| Day           | Time                  | Mode                                  | Required study and application                                                                                                                    |
|---------------|-----------------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon<br>14 Dec | 3 h                   | Process variables                     | Study focus: Process variables. Application: Identify controlled, measured and manipulated variables for a tank.                                  |
| Tue<br>15 Dec | 3 h                   | PV and SP                             | Study focus: PV and SP. Application: Calculate SP-PV error and state the assumed control direction.                                               |
| Wed<br>16 Dec | 3 h                   | Manipulated variable and disturbances | Study focus: Manipulated variable and disturbances. Application: Separate controller output from process response and disturbances.               |
| Thu<br>17 Dec | 3 h                   | Feedback control loops                | Study focus: Feedback control loops. Application: Draw process-sensor-controller-actuator feedback.                                               |
| Fri<br>18 Dec | 3 h                   | Manual and Automatic control          | Study focus: Manual and Automatic control. Application: Design a bumpless Manual-to-Auto transfer check at fundamentals level.                    |
| Sat<br>19 Dec | 4 h                   | Project 2 focus                       | Focus on Project 2 only for 4 hours. Choose the next incomplete project section from your own design/build plan; no separate Saturday laboratory. |
| Sun<br>20 Dec | Optional<br>30-60 min | Light consolidation or rest           | No test and no new topic. If useful, review the error log, organize evidence and preview Monday; otherwise rest.                                  |

### SESSION STRUCTURE

Monday-Friday: 15 min retrieval, 65 min focused learning, 10 min break, 45 min worked examples, 10 min break, 25 min application, 10 min error log and next action. Saturday: 4 hours using the laboratory or active-project rule shown above.

# M4 Module Week 2 - Programme Week 16

| WEEKLY FOCUS     |                                                                    |
|------------------|--------------------------------------------------------------------|
| Dates            | 21 Dec - 27 Dec 2026                                               |
| Theme            | Process Control II                                                 |
| Primary scope    | PID fundamentals, alarms, permissives, interlocks and ESD concepts |
| Portfolio window | Project 2 active - Saturday is project focus only                  |

## Weekly learning outcomes

[ ] Explain PID terms

[ ] Interpret trends

[ ] Design protective responses

## Monday-Sunday execution plan

| Day           | Time                  | Mode                                            | Required study and application                                                                                                                    |
|---------------|-----------------------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon<br>21 Dec | 3 h                   | Proportional action                             | Study focus: Proportional action. Application: Predict the directional effect of increasing proportional action.                                  |
| Tue<br>22 Dec | 3 h                   | Integral and derivative awareness               | Study focus: Integral and derivative awareness. Application: Explain integral removal of offset and why derivative reacts to rate of change.      |
| Wed<br>23 Dec | 3 h                   | Process response and PID trends                 | Study focus: Process response and PID trends. Application: Classify stable, slow, overshooting, oscillating and saturated trends.                 |
| Thu<br>24 Dec | 3 h                   | Alarm design and deadband                       | Study focus: Alarm design and deadband. Application: Create alarm priority, deadband and operator-response entries.                               |
| Fri<br>25 Dec | 3 h                   | Interlocks, permissives, trips and ESD concepts | Study focus: Interlocks, permissives, trips and ESD concepts. Application: Build a cause/effect row for high-high level trip and safe recovery.   |
| Sat<br>26 Dec | 4 h                   | Project 2 focus                                 | Focus on Project 2 only for 4 hours. Choose the next incomplete project section from your own design/build plan; no separate Saturday laboratory. |
| Sun<br>27 Dec | Optional<br>30-60 min | Light consolidation or rest                     | No test and no new topic. If useful, review the error log, organize evidence and preview Monday; otherwise rest.                                  |

### SESSION STRUCTURE

Monday-Friday: 15 min retrieval, 65 min focused learning, 10 min break, 45 min worked examples, 10 min break, 25 min application, 10 min error log and next action. Saturday: 4 hours using the laboratory or active-project rule shown above.

# M4 applied technician case study

## Tank approaches high-high level

### SITUATION

The level PV rises while the normal control output is saturated. High-high protection must act independently of routine control and recovery must require a healthy condition.

### Available evidence

- ☐ PV/SP/MV trend
- ☐ Alarm/trip state
- ☐ Permissive/interlock state
- ☐ Actuator command and feedback

### Technician reasoning tasks

- ☐ Separate control failure from protection
- ☐ Define safe cause/effect
- ☐ State operator information
- ☐ Prove recovery cannot bypass the trip

### Required case-study record

| Record section              | What to capture                                                |
|-----------------------------|----------------------------------------------------------------|
| Symptom                     | Observable behavior without premature diagnosis                |
| Initial checks              | Safe, non-invasive checks in logical order                     |
| Evidence                    | Readings, states, screenshots or traces                        |
| Isolation                   | Boundary where expected and actual first differ                |
| Root cause                  | Verified cause, not a guess                                    |
| Correction and verification | Controlled change, normal/fault/recovery retest and limitation |

### LEARNING USE

This case study is an applied comprehension activity, not an examination. Use references, correct misconceptions and preserve the reasoning trail.

# M5 - SCADA / HMI

| TARGET LEVEL | DATES                     | PLANNED CORE LOAD |
|--------------|---------------------------|-------------------|
| Developing   | 28 Dec 2026 - 10 Jan 2027 | 38 hours          |

## MODULE PURPOSE

Create operator displays that communicate state, ownership, alarm, quality, history and process response without hiding PLC authority.

## Module competencies

- ☐ Create screen hierarchy
- ☐ Show state and ownership
- ☐ Trace communications
- ☐ Design useful alarms
- ☐ Interpret trends/history
- ☐ Navigate Ignition fundamentals

## Module activities

- ☐ Design hierarchy and faceplates
- ☐ Configure or simulate tags
- ☐ Build alarms and trends
- ☐ Test stale/bad-quality behavior

## Weekly sequence

| Programme week | Dates           | Theme        | Primary scope                                                           |
|----------------|-----------------|--------------|-------------------------------------------------------------------------|
| Week 17        | 28 Dec - 03 Jan | SCADA/HMI I  | HMI/SCADA fundamentals, visualization, hierarchy and PLC communications |
| Week 18        | 04 Jan - 10 Jan | SCADA/HMI II | Alarm management, trends, history, historian concepts and Ignition      |

# M5 Module Week 1 - Programme Week 17

| WEEKLY FOCUS     |                                                                         |
|------------------|-------------------------------------------------------------------------|
| Dates            | 28 Dec - 03 Jan 2027                                                    |
| Theme            | SCADA/HMI I                                                             |
| Primary scope    | HMI/SCADA fundamentals, visualization, hierarchy and PLC communications |
| Portfolio window | Project 3 active - Saturday is project focus only                       |

## Weekly learning outcomes

- [ ] Create screen hierarchy
- [ ] Show state and ownership
- [ ] Trace communications

## Monday-Sunday execution plan

| Day           | Time                  | Mode                                     | Required study and application                                                                                                                             |
|---------------|-----------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon<br>28 Dec | 3 h                   | HMI versus SCADA                         | Study focus: HMI versus SCADA. Application: Compare local HMI and supervisory SCADA roles.                                                                 |
| Tue<br>29 Dec | 3 h                   | Process visualization principles         | Study focus: Process visualization principles. Application: Redesign a cluttered screen using neutral normal states.                                       |
| Wed<br>30 Dec | 3 h                   | Screen hierarchy and navigation          | Study focus: Screen hierarchy and navigation. Application: Create overview, area, equipment, alarm and trend navigation.                                   |
| Thu<br>31 Dec | 3 h                   | Equipment faceplates                     | Study focus: Equipment faceplates. Application: Design a pump faceplate showing command, mode, feedback, permissives and fault.                            |
| Fri<br>01 Jan | 3 h                   | PLC communications and command ownership | Study focus: PLC communications and command ownership. Application: Trace an HMI command through communications and PLC ownership logic.                   |
| Sat<br>02 Jan | 4 h                   | Project 3 focus                          | Focus on Project 3 only for 4 hours. Choose the next incomplete flagship-project section from your own design/build plan; no separate Saturday laboratory. |
| Sun<br>03 Jan | Optional<br>30-60 min | Light consolidation or rest              | No test and no new topic. If useful, review the error log, organize evidence and preview Monday; otherwise rest.                                           |

### SESSION STRUCTURE

Monday-Friday: 15 min retrieval, 65 min focused learning, 10 min break, 45 min worked examples, 10 min break, 25 min application, 10 min error log and next action. Saturday: 4 hours using the laboratory or active-project rule shown above.

# M5 Module Week 2 - Programme Week 18

| WEEKLY FOCUS     |                                                                    |
|------------------|--------------------------------------------------------------------|
| Dates            | 04 Jan - 10 Jan 2027                                               |
| Theme            | SCADA/HMI II                                                       |
| Primary scope    | Alarm management, trends, history, historian concepts and Ignition |
| Portfolio window | Project 3 active - Saturday is project focus only                  |

## Weekly learning outcomes

- [ ] Design useful alarms
- [ ] Interpret trends/history
- [ ] Navigate Ignition fundamentals

## Monday-Sunday execution plan

| Day           | Time                  | Mode                                   | Required study and application                                                                                                                             |
|---------------|-----------------------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon<br>04 Jan | 3 h                   | Alarm philosophy                       | Study focus: Alarm philosophy. Application: Separate alarm condition, priority, message, response and shelving awareness.                                  |
| Tue<br>05 Jan | 3 h                   | Alarm priorities and acknowledgement   | Study focus: Alarm priorities and acknowledgement. Application: Explain acknowledge versus return-to-normal and avoid nuisance alarms.                     |
| Wed<br>06 Jan | 3 h                   | Trends                                 | Study focus: Trends. Application: Select pens, scales and time windows for a tank diagnostic trend.                                                        |
| Thu<br>07 Jan | 3 h                   | Historical data and historian concepts | Study focus: Historical data and historian concepts. Application: Distinguish live, historical, stale and bad-quality data.                                |
| Fri<br>08 Jan | 3 h                   | Ignition fundamentals                  | Study focus: Ignition fundamentals. Application: Build or plan an Ignition tag, alarm, trend and Perspective view workflow.                                |
| Sat<br>09 Jan | 4 h                   | Project 3 focus                        | Focus on Project 3 only for 4 hours. Choose the next incomplete flagship-project section from your own design/build plan; no separate Saturday laboratory. |
| Sun<br>10 Jan | Optional<br>30-60 min | Light consolidation or rest            | No test and no new topic. If useful, review the error log, organize evidence and preview Monday; otherwise rest.                                           |

### SESSION STRUCTURE

Monday-Friday: 15 min retrieval, 65 min focused learning, 10 min break, 45 min worked examples, 10 min break, 25 min application, 10 min error log and next action. Saturday: 4 hours using the laboratory or active-project rule shown above.

# M5 applied technician case study

## A normal-looking level remains after communications fail

### SITUATION

The screen continues showing the last plausible value even though the controller connection has failed. The operator cannot distinguish live from stale data.

### Available evidence

- ☐ Tag quality
- ☐ Timestamp
- ☐ Communication state
- ☐ Alarm and display behavior

### Technician reasoning tasks

- ☐ Identify the misleading presentation
- ☐ Design bad-quality behavior
- ☐ Prevent unsafe command assumptions
- ☐ Verify loss and recovery

### Required case-study record

| Record section              | What to capture                                                |
|-----------------------------|----------------------------------------------------------------|
| Symptom                     | Observable behavior without premature diagnosis                |
| Initial checks              | Safe, non-invasive checks in logical order                     |
| Evidence                    | Readings, states, screenshots or traces                        |
| Isolation                   | Boundary where expected and actual first differ                |
| Root cause                  | Verified cause, not a guess                                    |
| Correction and verification | Controlled change, normal/fault/recovery retest and limitation |

### LEARNING USE

This case study is an applied comprehension activity, not an examination. Use references, correct misconceptions and preserve the reasoning trail.

# M6 - Industrial Networking

| TARGET LEVEL      | DATES                     | PLANNED CORE LOAD |
|-------------------|---------------------------|-------------------|
| Working knowledge | 11 Jan 2027 - 24 Jan 2027 | 38 hours          |

## MODULE PURPOSE

Develop Ethernet, addressing, protocol and layered troubleshooting skills for PLC/HMI/SCADA communications.

## Module competencies

- ☐ Draw network topology
- ☐ Create an IP plan
- ☐ Troubleshoot by layer
- ☐ Explain common protocols
- ☐ Record serial settings
- ☐ Diagnose network faults

## Module activities

- ☐ Create topology and IP schedules
- ☐ Document serial settings
- ☐ Map Modbus/OPC data
- ☐ Inject safe communication faults

## Weekly sequence

| Programme week | Dates           | Theme         | Primary scope                                                       |
|----------------|-----------------|---------------|---------------------------------------------------------------------|
| Week 19        | 11 Jan - 17 Jan | Networking I  | Ethernet, topology, IP addressing, subnetting and TCP/IP            |
| Week 20        | 18 Jan - 24 Jan | Networking II | Modbus, OPC UA, serial communications, Profinet and troubleshooting |

# M6 Module Week 1 - Programme Week 19

| WEEKLY FOCUS     |                                                          |
|------------------|----------------------------------------------------------|
| Dates            | 11 Jan - 17 Jan 2027                                     |
| Theme            | Networking I                                             |
| Primary scope    | Ethernet, topology, IP addressing, subnetting and TCP/IP |
| Portfolio window | Project 3 active - Saturday is project focus only        |

## Weekly learning outcomes

[ ] Draw network topology

[ ] Create an IP plan

[ ] Troubleshoot by layer

## Monday-Sunday execution plan

| Day           | Time                  | Mode                        | Required study and application                                                                                                                             |
|---------------|-----------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon<br>11 Jan | 3 h                   | Ethernet fundamentals       | Study focus: Ethernet fundamentals. Application: Draw PLC, HMI, SCADA and device Ethernet links.                                                           |
| Tue<br>12 Jan | 3 h                   | Industrial network topology | Study focus: Industrial network topology. Application: Compare a star topology with a small redundant concept at awareness level.                          |
| Wed<br>13 Jan | 3 h                   | IPv4 addressing             | Study focus: IPv4 addressing. Application: Assign unique addresses and create an IP schedule.                                                              |
| Thu<br>14 Jan | 3 h                   | Subnet masks and gateways   | Study focus: Subnet masks and gateways. Application: Determine whether two /24 addresses are local peers.                                                  |
| Fri<br>15 Jan | 3 h                   | TCP/IP, ports and ping      | Study focus: TCP/IP, ports and ping. Application: Explain why ping success does not prove an application service is healthy.                               |
| Sat<br>16 Jan | 4 h                   | Project 3 focus             | Focus on Project 3 only for 4 hours. Choose the next incomplete flagship-project section from your own design/build plan; no separate Saturday laboratory. |
| Sun<br>17 Jan | Optional<br>30-60 min | Light consolidation or rest | No test and no new topic. If useful, review the error log, organize evidence and preview Monday; otherwise rest.                                           |

### SESSION STRUCTURE

Monday-Friday: 15 min retrieval, 65 min focused learning, 10 min break, 45 min worked examples, 10 min break, 25 min application, 10 min error log and next action. Saturday: 4 hours using the laboratory or active-project rule shown above.

# M6 Module Week 2 - Programme Week 20

| WEEKLY FOCUS     |                                                                     |
|------------------|---------------------------------------------------------------------|
| Dates            | 18 Jan - 24 Jan 2027                                                |
| Theme            | Networking II                                                       |
| Primary scope    | Modbus, OPC UA, serial communications, Profinet and troubleshooting |
| Portfolio window | Project 3 active - Saturday is project focus only                   |

## Weekly learning outcomes

[ ] Explain common protocols

[ ] Record serial settings

[ ] Diagnose network faults

## Monday-Sunday execution plan

| Day           | Time                  | Mode                               | Required study and application                                                                                                                             |
|---------------|-----------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon<br>18 Jan | 3 h                   | Modbus TCP fundamentals            | Study focus: Modbus TCP fundamentals. Application: Map device ID, function, register, datatype and byte order.                                             |
| Tue<br>19 Jan | 3 h                   | Modbus RTU and serial settings     | Study focus: Modbus RTU and serial settings. Application: Match baud, parity, stop bits, node address and termination.                                     |
| Wed<br>20 Jan | 3 h                   | OPC UA fundamentals                | Study focus: OPC UA fundamentals. Application: Explain OPC UA client/server, tags and quality at fundamentals level.                                       |
| Thu<br>21 Jan | 3 h                   | Profinet awareness                 | Study focus: Profinet awareness. Application: Compare Profinet device naming and IP awareness with ordinary Ethernet.                                      |
| Fri<br>22 Jan | 3 h                   | Industrial network troubleshooting | Study focus: Industrial network troubleshooting. Application: Diagnose link, address, transport, protocol, mapping and application layers in order.        |
| Sat<br>23 Jan | 4 h                   | Project 3 focus                    | Focus on Project 3 only for 4 hours. Choose the next incomplete flagship-project section from your own design/build plan; no separate Saturday laboratory. |
| Sun<br>24 Jan | Optional<br>30-60 min | Light consolidation or rest        | No test and no new topic. If useful, review the error log, organize evidence and preview Monday; otherwise rest.                                           |

### SESSION STRUCTURE

Monday-Friday: 15 min retrieval, 65 min focused learning, 10 min break, 45 min worked examples, 10 min break, 25 min application, 10 min error log and next action. Saturday: 4 hours using the laboratory or active-project rule shown above.

# M6 applied technician case study

## HMI stopped communicating after maintenance

### SITUATION

The physical link is present, but the HMI cannot read the controller after an address change. The learner must test physical, addressing, transport, protocol and application layers.

### Available evidence

- ☐ Link state
- ☐ IP/subnet results
- ☐ Port/service state
- ☐ Driver and tag diagnostics

### Technician reasoning tasks

- ☐ Start at the physical layer
- ☐ Confirm unique compatible addresses
- ☐ Check service/protocol mapping
- ☐ Document correction and reconnection

### Required case-study record

| Record section              | What to capture                                                |
|-----------------------------|----------------------------------------------------------------|
| Symptom                     | Observable behavior without premature diagnosis                |
| Initial checks              | Safe, non-invasive checks in logical order                     |
| Evidence                    | Readings, states, screenshots or traces                        |
| Isolation                   | Boundary where expected and actual first differ                |
| Root cause                  | Verified cause, not a guess                                    |
| Correction and verification | Controlled change, normal/fault/recovery retest and limitation |

### LEARNING USE

This case study is an applied comprehension activity, not an examination. Use references, correct misconceptions and preserve the reasoning trail.

# M7 - Engineering Documentation

| TARGET LEVEL | DATES                     | PLANNED CORE LOAD |
|--------------|---------------------------|-------------------|
| Developing   | 25 Jan 2027 - 07 Feb 2027 | 38 hours          |

## MODULE PURPOSE

Produce coordinated drawings, registers, narratives, matrices, test procedures and fault reports that another technician can follow.

## Module competencies

- ☐ Trace drawings
- ☐ Coordinate tags
- ☐ Build usable registers
- ☐ Write maintainable documents
- ☐ Create traceable matrices
- ☐ Report troubleshooting clearly

## Module activities

- ☐ Create a P&ID and schematics
- ☐ Coordinate I/O/instrument lists
- ☐ Write narratives and matrices
- ☐ Cross-check documents against logic

## Weekly sequence

| Programme week | Dates           | Theme            | Primary scope                                                        |
|----------------|-----------------|------------------|----------------------------------------------------------------------|
| Week 21        | 25 Jan - 31 Jan | Documentation I  | P&IDs, schematics, wiring/loop diagrams and registers                |
| Week 22        | 01 Feb - 07 Feb | Documentation II | Narratives, matrices, alarm/network documents and technician reports |

# M7 Module Week 1 - Programme Week 21

| WEEKLY FOCUS     |                                                       |
|------------------|-------------------------------------------------------|
| Dates            | 25 Jan - 31 Jan 2027                                  |
| Theme            | Documentation I                                       |
| Primary scope    | P&IDs, schematics, wiring/loop diagrams and registers |
| Portfolio window | Project 3 active - Saturday is project focus only     |

## Weekly learning outcomes

- [ ] Trace drawings
- [ ] Coordinate tags
- [ ] Build usable registers

## Monday-Sunday execution plan

| Day           | Time                  | Mode                        | Required study and application                                                                                                                             |
|---------------|-----------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon<br>25 Jan | 3 h                   | P&IDs                       | Study focus: P&IDs. Application: Trace equipment, piping, valves and an instrument loop on a tank P&ID.                                                    |
| Tue<br>26 Jan | 3 h                   | Electrical schematics       | Study focus: Electrical schematics. Application: Trace protection, command, coil and feedback on a motor schematic.                                        |
| Wed<br>27 Jan | 3 h                   | Wiring diagrams             | Study focus: Wiring diagrams. Application: Create terminal-to-terminal wiring for a simulated transmitter.                                                 |
| Thu<br>28 Jan | 3 h                   | Instrument loop diagrams    | Study focus: Instrument loop diagrams. Application: Draw process-to-HMI loop boundaries and safe verification points.                                      |
| Fri<br>29 Jan | 3 h                   | I/O and instrument lists    | Study focus: I/O and instrument lists. Application: Coordinate tags, ranges, addresses, normal/fail states and descriptions.                               |
| Sat<br>30 Jan | 4 h                   | Project 3 focus             | Focus on Project 3 only for 4 hours. Choose the next incomplete flagship-project section from your own design/build plan; no separate Saturday laboratory. |
| Sun<br>31 Jan | Optional<br>30-60 min | Light consolidation or rest | No test and no new topic. If useful, review the error log, organize evidence and preview Monday; otherwise rest.                                           |

### SESSION STRUCTURE

Monday-Friday: 15 min retrieval, 65 min focused learning, 10 min break, 45 min worked examples, 10 min break, 25 min application, 10 min error log and next action. Saturday: 4 hours using the laboratory or active-project rule shown above.

# M7 Module Week 2 - Programme Week 22

| WEEKLY FOCUS     |                                                                      |
|------------------|----------------------------------------------------------------------|
| Dates            | 01 Feb - 07 Feb 2027                                                 |
| Theme            | Documentation II                                                     |
| Primary scope    | Narratives, matrices, alarm/network documents and technician reports |
| Portfolio window | Project 3 active - Saturday is project focus only                    |

## Weekly learning outcomes

[ ] Write maintainable documents

[ ] Create traceable matrices

[ ] Report troubleshooting clearly

## Monday-Sunday execution plan

| Day           | Time                  | Mode                                        | Required study and application                                                                                                                             |
|---------------|-----------------------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon<br>01 Feb | 3 h                   | Control narratives                          | Study focus: Control narratives. Application: Write purpose, modes, sequence, protections and recovery in a control narrative.                             |
| Tue<br>02 Feb | 3 h                   | Sequence descriptions                       | Study focus: Sequence descriptions. Application: Define steps and transitions without ambiguous words.                                                     |
| Wed<br>03 Feb | 3 h                   | Cause/effect and trip/interlock matrices    | Study focus: Cause/effect and trip/interlock matrices. Application: Create traceable cause/effect and trip/interlock rows.                                 |
| Thu<br>04 Feb | 3 h                   | Alarm lists and network diagrams            | Study focus: Alarm lists and network diagrams. Application: Coordinate alarm tags with a labelled network drawing.                                         |
| Fri<br>05 Feb | 3 h                   | Test procedures and troubleshooting reports | Study focus: Test procedures and troubleshooting reports. Application: Write expected result, actual result, evidence, defect and retest fields.           |
| Sat<br>06 Feb | 4 h                   | Project 3 focus                             | Focus on Project 3 only for 4 hours. Choose the next incomplete flagship-project section from your own design/build plan; no separate Saturday laboratory. |
| Sun<br>07 Feb | Optional<br>30-60 min | Light consolidation or rest                 | No test and no new topic. If useful, review the error log, organize evidence and preview Monday; otherwise rest.                                           |

### SESSION STRUCTURE

Monday-Friday: 15 min retrieval, 65 min focused learning, 10 min break, 45 min worked examples, 10 min break, 25 min application, 10 min error log and next action. Saturday: 4 hours using the laboratory or active-project rule shown above.

# M7 applied technician case study

## Pump-skid modification has conflicting documents

### SITUATION

The P&ID, I/O list, PLC tag and HMI label use different identifiers for the same device after a change.

### Available evidence

- ☐ Document revisions
- ☐ Tag cross-reference
- ☐ PLC/HMI evidence
- ☐ Approved change note

### Technician reasoning tasks

- ☐ Identify the controlling baseline
- ☐ Resolve every mismatch
- ☐ Update affected documents
- ☐ Record verification and limitations

### Required case-study record

| Record section              | What to capture                                                |
|-----------------------------|----------------------------------------------------------------|
| Symptom                     | Observable behavior without premature diagnosis                |
| Initial checks              | Safe, non-invasive checks in logical order                     |
| Evidence                    | Readings, states, screenshots or traces                        |
| Isolation                   | Boundary where expected and actual first differ                |
| Root cause                  | Verified cause, not a guess                                    |
| Correction and verification | Controlled change, normal/fault/recovery retest and limitation |

### LEARNING USE

This case study is an applied comprehension activity, not an examination. Use references, correct misconceptions and preserve the reasoning trail.

# M8 - Safety Awareness, Commissioning & Troubleshooting

| TARGET LEVEL                  | DATES                     | PLANNED CORE LOAD |
|-------------------------------|---------------------------|-------------------|
| Fundamentals with integration | 08 Feb 2027 - 28 Feb 2027 | 57 hours          |

## MODULE PURPOSE

Apply safety boundaries, commissioning discipline, FAT/SAT-style checks and structured fault-finding to the Project 3 flagship simulator.

## Module competencies

- ☐ Identify hazards
- ☐ Respect authorization limits
- ☐ Classify protective functions
- ☐ Plan commissioning
- ☐ Execute traceable checks
- ☐ Control defects and restoration
- ☐ Troubleshoot systematically
- ☐ Verify correction

## Module activities

- ☐ Prepare hazard and boundary notes
- ☐ Execute I/O/loop checks
- ☐ Manage punch items
- ☐ Run integrated faults and regression

## Weekly sequence

| Programme week | Dates           | Theme                                   | Primary scope                                                       |
|----------------|-----------------|-----------------------------------------|---------------------------------------------------------------------|
| Week 23        | 08 Feb - 14 Feb | Safety Awareness                        | LOTO, permits, electrical hazards, emergency stops and ESD concepts |
| Week 24        | 15 Feb - 21 Feb | Commissioning                           | Pre-commissioning, I/O checks, loop checks, FAT/SAT and punch lists |
| Week 25        | 22 Feb - 28 Feb | Integrated Troubleshooting and Handover | Structured fault-finding, regression, demonstration and release     |

# M8 Module Week 1 - Programme Week 23

| WEEKLY FOCUS     |                                                                     |
|------------------|---------------------------------------------------------------------|
| Dates            | 08 Feb - 14 Feb 2027                                                |
| Theme            | Safety Awareness                                                    |
| Primary scope    | LOTO, permits, electrical hazards, emergency stops and ESD concepts |
| Portfolio window | Project 3 active - Saturday is project focus only                   |

## Weekly learning outcomes

[ ] Identify hazards

[ ] Respect authorization limits

[ ] Classify protective functions

## Monday-Sunday execution plan

| Day           | Time                  | Mode                                             | Required study and application                                                                                                                             |
|---------------|-----------------------|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon<br>08 Feb | 3 h                   | Industrial hazard identification                 | Study focus: Industrial hazard identification. Application: Identify electrical, stored-energy, rotating, pressure and water/electronics hazards.          |
| Tue<br>09 Feb | 3 h                   | Lockout/Tagout awareness                         | Study focus: Lockout/Tagout awareness. Application: Explain isolation, lock, tag, verification and authorization boundaries.                               |
| Wed<br>10 Feb | 3 h                   | Permit-to-work awareness                         | Study focus: Permit-to-work awareness. Application: List permit purpose, limits, responsible people and close-out.                                         |
| Thu<br>11 Feb | 3 h                   | Electrical safety and test-before-touch          | Study focus: Electrical safety and test-before-touch. Application: Create a safe test plan for a low-voltage trainer and state stop-work triggers.         |
| Fri<br>12 Feb | 3 h                   | Emergency-stop, ESD and hazardous-area awareness | Study focus: Emergency-stop, ESD and hazardous-area awareness. Application: Differentiate a training E-stop input from a certified safety function.        |
| Sat<br>13 Feb | 4 h                   | Project 3 focus                                  | Focus on Project 3 only for 4 hours. Choose the next incomplete flagship-project section from your own design/build plan; no separate Saturday laboratory. |
| Sun<br>14 Feb | Optional<br>30-60 min | Light consolidation or rest                      | No test and no new topic. If useful, review the error log, organize evidence and preview Monday; otherwise rest.                                           |

### SESSION STRUCTURE

Monday-Friday: 15 min retrieval, 65 min focused learning, 10 min break, 45 min worked examples, 10 min break, 25 min application, 10 min error log and next action. Saturday: 4 hours using the laboratory or active-project rule shown above.

# M8 Module Week 2 - Programme Week 24

| WEEKLY FOCUS     |                                                                     |
|------------------|---------------------------------------------------------------------|
| Dates            | 15 Feb - 21 Feb 2027                                                |
| Theme            | Commissioning                                                       |
| Primary scope    | Pre-commissioning, I/O checks, loop checks, FAT/SAT and punch lists |
| Portfolio window | Project 3 active - Saturday is project focus only                   |

## Weekly learning outcomes

- [ ] Plan commissioning
- [ ] Execute traceable checks
- [ ] Control defects and restoration

## Monday-Sunday execution plan

| Day           | Time                  | Mode                                            | Required study and application                                                                                                                             |
|---------------|-----------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon<br>15 Feb | 3 h                   | Commissioning planning                          | Study focus: Commissioning planning. Application: Define scope, roles, prerequisites, hold points and safe boundaries.                                     |
| Tue<br>16 Feb | 3 h                   | Visual and cold checks                          | Study focus: Visual and cold checks. Application: Inspect labels, polarity, protection, termination and document revision before power.                    |
| Wed<br>17 Feb | 3 h                   | I/O and loop checks                             | Study focus: I/O and loop checks. Application: Prove input stimulus, PLC state, output command, feedback and HMI indication.                               |
| Thu<br>18 Feb | 3 h                   | FAT/SAT-style testing                           | Study focus: FAT/SAT-style testing. Application: Separate factory-style and site-style acceptance evidence for a home simulation.                          |
| Fri<br>19 Feb | 3 h                   | Defects, punch lists and controlled restoration | Study focus: Defects, punch lists and controlled restoration. Application: Classify, correct, retest and close defects with traceability.                  |
| Sat<br>20 Feb | 4 h                   | Project 3 focus                                 | Focus on Project 3 only for 4 hours. Choose the next incomplete flagship-project section from your own design/build plan; no separate Saturday laboratory. |
| Sun<br>21 Feb | Optional<br>30-60 min | Light consolidation or rest                     | No test and no new topic. If useful, review the error log, organize evidence and preview Monday; otherwise rest.                                           |

### SESSION STRUCTURE

Monday-Friday: 15 min retrieval, 65 min focused learning, 10 min break, 45 min worked examples, 10 min break, 25 min application, 10 min error log and next action. Saturday: 4 hours using the laboratory or active-project rule shown above.

# M8 Module Week 3 - Programme Week 25

| WEEKLY FOCUS     |                                                                 |
|------------------|-----------------------------------------------------------------|
| Dates            | 22 Feb - 28 Feb 2027                                            |
| Theme            | Integrated Troubleshooting and Handover                         |
| Primary scope    | Structured fault-finding, regression, demonstration and release |
| Portfolio window | Project 3 active - Saturday is project focus only               |

## Weekly learning outcomes

- [ ] Troubleshoot systematically
- [ ] Verify correction
- [ ] Deliver an honest evidence pack

## Monday-Sunday execution plan

| Day           | Time                  | Mode                                             | Required study and application                                                                                                                              |
|---------------|-----------------------|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon<br>22 Feb | 3 h                   | Symptom-to-root-cause method                     | Study focus: Symptom-to-root-cause method. Application: Use symptom, checks, evidence, isolation, diagnosis, correction and verification in order.          |
| Tue<br>23 Feb | 3 h                   | Electrical and I/O fault isolation               | Study focus: Electrical and I/O fault isolation. Application: Trace source, protection, field input, I/O, logic, interface and output.                      |
| Wed<br>24 Feb | 3 h                   | Instrument/process fault isolation               | Study focus: Instrument/process fault isolation. Application: Separate bad measurement from real process change.                                            |
| Thu<br>25 Feb | 3 h                   | HMI/network fault isolation                      | Study focus: HMI/network fault isolation. Application: Diagnose quality, connectivity, address, protocol, tag mapping and ownership.                        |
| Fri<br>26 Feb | 3 h                   | Regression, handover and technical demonstration | Study focus: Regression, handover and technical demonstration. Application: Run regression, back up files and explain the flagship project in 5-10 minutes. |
| Sat<br>27 Feb | 4 h                   | Project 3 focus                                  | Focus on Project 3 only for 4 hours. Choose the next incomplete flagship-project section from your own design/build plan; no separate Saturday laboratory.  |
| Sun<br>28 Feb | Optional<br>30-60 min | Light consolidation or rest                      | No test and no new topic. If useful, review the error log, organize evidence and preview Monday; otherwise rest.                                            |

### SESSION STRUCTURE

Monday-Friday: 15 min retrieval, 65 min focused learning, 10 min break, 45 min worked examples, 10 min break, 25 min application, 10 min error log and next action. Saturday: 4 hours using the laboratory or active-project rule shown above.

# M8 applied technician case study

## Instrument fault during simulated operation

### SITUATION

A process value suddenly becomes implausible while the pump is commanded. The learner must protect the simulated process, distinguish real change from measurement failure and restore only after verification.

### Available evidence

- ☐ Alarm/trip chronology
- ☐ Raw and engineering values
- ☐ I/O/loop checks
- ☐ Correction and regression results

### Technician reasoning tasks

- ☐ State the safe immediate response
- ☐ Trace sensor to HMI
- ☐ Diagnose without bypassing protection
- ☐ Verify restoration and document the lesson

### Required case-study record

| Record section              | What to capture                                                |
|-----------------------------|----------------------------------------------------------------|
| Symptom                     | Observable behavior without premature diagnosis                |
| Initial checks              | Safe, non-invasive checks in logical order                     |
| Evidence                    | Readings, states, screenshots or traces                        |
| Isolation                   | Boundary where expected and actual first differ                |
| Root cause                  | Verified cause, not a guess                                    |
| Correction and verification | Controlled change, normal/fault/recovery retest and limitation |

### LEARNING USE

This case study is an applied comprehension activity, not an examination. Use references, correct misconceptions and preserve the reasoning trail.

# Programme completion standard

Treat the syllabus as complete when the following can be demonstrated honestly through simulation, safe hobby-scale hardware and documented project evidence.

| TECHNICAL CAPABILITY                                        | PROFESSIONAL EVIDENCE                                       |
|-------------------------------------------------------------|-------------------------------------------------------------|
| <input type="checkbox"/> Trace electrical and control paths | <input type="checkbox"/> Exactly three integrated projects  |
| <input type="checkbox"/> Write/test Ladder Logic            | <input type="checkbox"/> Coordinated engineering documents  |
| <input type="checkbox"/> Scale and diagnose analog signals  | <input type="checkbox"/> Normal/fault/recovery results      |
| <input type="checkbox"/> Explain PV/SP/PID fundamentals     | <input type="checkbox"/> Structured troubleshooting reports |
| <input type="checkbox"/> Build HMI alarms and trends        | <input type="checkbox"/> FAT/SAT-style records              |
| <input type="checkbox"/> Diagnose basic network faults      | <input type="checkbox"/> Clear real/simulated/DIY labels    |

## FINAL STANDARD

Competence is shown by reasoning, working behavior, traceability and honest limitations. The programme contains no Sunday examinations; readiness grows through repeated application, project work and corrected evidence.